

5.2.3 Average Value

Motivation: How do you compute the average of four numbers: a , b , c , & d ?

Now suppose that we wish to compute the average y -value of a function. Using the previous idea to compute such an average is much more difficult, because most functions of interest have infinitely many y -values. So, how would approach such a question?

Definition: The **average value** of f on $[a, b]$ is:

$$f_{\text{avg}} = \frac{1}{b-a} \int_a^b f(x) dx$$

Example 1: Find the average value of the function $f(x) = 1 + x$ on the interval $[0, 3]$.

Example 2: Find the average value of the function $f(x) = \sqrt{1-x^2}$ on the interval $[-1, 1]$.

Question: If we are always finding the average value of a continuous function, must it be the case that there is a value of x where $f(x) = f_{\text{avg}}$?

Theorem (The Mean Value Theorem for Integrals): If f is continuous on $[a, b]$, then there exists a number c in $[a, b]$ such that:

$$f(c) = f_{\text{avg}} = \frac{1}{b-a} \int_a^b f(x) dx$$

Proof:

Example 3: Find all numbers c that satisfy the MVT for Integrals for the function $f(x) = 1 + x$ on the interval $[0, 3]$.

Example 4: Find all numbers c that satisfy the MVT for Integrals for the function $f(x) = \sqrt{1-x^2}$ on the interval $[-1, 1]$.

Example 5: If f is a continuous function and $\int_2^7 f(x) dx = \frac{11}{2}$, show that f takes on the value of 1.1 at least once on the interval $[2, 7]$.

Note: The Average Value of a function and the MVT for Integrals are powerful concepts that will help us to connect the various ideas of Calculus in what we call the Fundamental Theorem of Calculus!!!